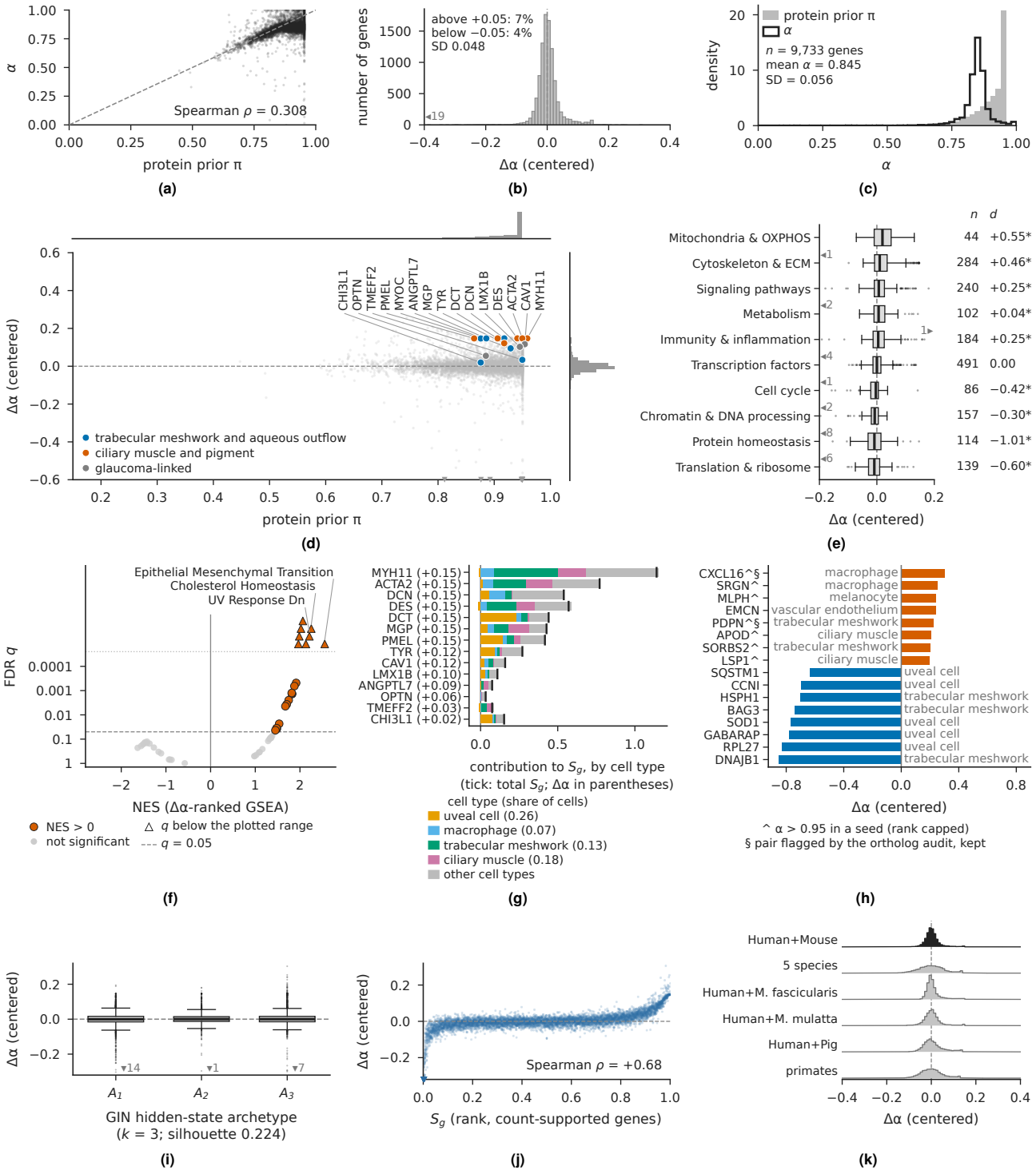

Spearman ρ = 0.308
protein prior π
α
(a)
above +0.05: 7%
below −0.05: 4%
SD 0.048
number of genes
Δα (centered)
(b)
protein prior π
α
n = 9,733 genes
mean α = 0.845
SD = 0.056
density
(c)
CHI3L1
OPTN
TMEFF2
PMEL
MYOC
ANGPTL7
MGP
TYR
DCT
DCN
LMX1B
DES
ACTA2
CAV1
MYH11
trabecular meshwork and aqueous outflow
ciliary muscle and pigment
glaucoma-linked
Δα (centered)
protein prior π
(d)
n
d
Mitochondria & OXPHOS 44 +0.55*
Cytoskeleton & ECM 284 +0.46*
Signaling pathways 240 +0.25*
Metabolism 102 +0.04*
Immunity & inflammation 184 +0.25*
Transcription factors 491 0.00
Cell cycle 86 −0.42*
Chromatin & DNA processing 157 −0.30*
Protein homeostasis 114 −1.01*
Translation & ribosome 139 −0.60*
Δα (centered)
(e)
Epithelial Mesenchymal Transition
Cholesterol Homeostasis
UV Response Dn
FDR q
NES (Δα-ranked GSEA)
NES > 0
not significant
q below the plotted range
q = 0.05
(f)
MYH11 (+0.15)
ACTA2 (+0.15)
DCN (+0.15)
DES (+0.15)
DCT (+0.15)
MGP (+0.15)
PMEL (+0.15)
TYR (+0.12)
CAV1 (+0.12)
LMX1B (+0.10)
ANGPTL7 (+0.09)
OPTN (+0.06)
TMEFF2 (+0.03)
CHI3L1 (+0.02)
contribution to S_g, by cell type
(tick: total S_g; Δα in parentheses)
cell type (share of cells)
uveal cell (0.26)
macrophage (0.07)
trabecular meshwork (0.13)
ciliary muscle (0.18)
other cell types
(g)
CXCL16^§ macrophage
SRGN^ macrophage
MLPH^ melanocyte
EMCN vascular endothelium
PDPN^§ trabecular meshwork
APOD^ ciliary muscle
SORBS2^ trabecular meshwork
LSP1^ ciliary muscle
SQSTM1 uveal cell
CCNI uveal cell
HSPH1 trabecular meshwork
BAG3 trabecular meshwork
SOD1 uveal cell
GABARAP uveal cell
RPL27 uveal cell
DNAJB1 trabecular meshwork
Δα (centered)
^ α > 0.95 in a seed (rank capped)
§ pair flagged by the ortholog audit, kept
(h)
Δα (centered)
GIN hidden-state archetype
(k = 3; silhouette 0.224)
(i)
Spearman ρ = +0.68
S_g (rank, count-supported genes)
Δα (centered)
(j)
Human+Mouse
5 species
Human+M. fascicularis
Human+M. mulatta
Human+Pig
primates
Δα (centered)
(k)